

Yejun (Henry) Kim

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Education

University of Virginia, Charlottesville, VA 2021–2025
Bachelor of Science, Aerospace Engineering & Mathematics
Cumulative GPA: 3.6 / 4.0
Honors: Dean's List, QuestBridge National College Match Scholar Relevant Coursework: Thermodynamics, Astronautics, Aerospace Structures, Aerospace Materials, Aerodynamics, Flight Vehicle Dynamics

Work Experience

Epic Systems | Technical Solutions Engineer Madison, WI | Jul 2025–Present

- Resolved high-severity production issues in medical scheduling software via backend M programming, data validation, and root-cause analysis across 200+ databases

Traffic Operations Laboratory | Undergraduate Researcher Charlottesville, VA | Spring 2025

- Applied time-series analysis, signal filtering, and control logic to real-time vehicular sensor data using Python and Raspberry Pi

We-Xite Lab | Undergraduate Researcher Storrs, CT | Summer 2024

- Fabricated a vibration isolation system and built a validated orientation control system for a submersible turbine, using 2D FFT analysis in MATLAB to characterize frequency response

Projects & Extracurriculars

Orbital Transfer Calculator Dec 2025

- Developed JavaScript-based visualization tools computing ΔV for Hohmann transfers, plane changes, and apse line maneuvers
- Conducted Izzo's method for Lambert's problem, Householder root-finding, and Nelder–Mead optimization

Underwater Robotics Team | Software Co-Lead Charlottesville, VA | 2024–2025

- Developed guidance logic to convert detected target position into heading/depth corrections for closed-loop vehicle control, coordinating with the hardware team on actuation response
- Processed real-time sonar/vision data to estimate relative position and orientation of the ROV to underwater targets

RASC-AL Capstone Project | Electrical and Controls Lead 2024–2025

- Derived operating voltage requirements for a parallel-plate electrostatic precipitator from particle kinematics and Paschen's Law, establishing a 1,030–8,910 V design window validated against simulated Martian atmospheric conditions (610 Pa, CO₂)
- Designed EP, airlock, and sensor control circuits and produced a system power budget (11.4 of 30 kWh allotted)

AquaMetric Automatic Water Dispensing System Fall 2024

- Modeled system dynamics from sensor input (pressure) to estimate a control variable (drain time) in real time, using regression-corrected theoretical models to reduce prediction error by 97% (MSE 10.2 → 0.297)

Skills

Languages: Python, MATLAB, R, C++, SQL

Tools: Git, Simulink, SolidWorks, ANSYS, Raspberry Pi, KiCad

Frameworks/Libraries: NumPy, pandas, scikit-learn, OpenCV